

Appendix of Measuring Individual User Fairness with User Similarity and Effectiveness Disparity

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Abstract. This is the appendix of the paper “Measuring Individual User Fairness with User Similarity and Effectiveness Disparity”, which has been accepted for presentation at the ECIR 2026 Full Papers track as a Findings paper.

Keywords: individual fairness · evaluation · recommender systems

A Measure Formulations

To improve reproducibility, we provide the formulations of all existing individual user fairness (FAIR) measures that we use.

A.1 Standard Deviation (SD)

SD measures how dispersed the data is in relation to the mean. Fairness can be measured by computing the SD of user utility, based on the user’s recommendation list [1, 7, 9]. The utility of user u , when given the top k recommendation list of user u' , $L_{u'}$ is originally defined as follows:

$$\phi_u^{ori}(L_{u'}) = \frac{\sum_{i \in L_{u'}} \hat{r}_{u,i}}{\sum_{i \in R_{u'}^k} \hat{r}_{u,i}} \quad (1)$$

where $\hat{r}_{u,i}$ is the predicted relevance score of user u for item i , and R_u^k is the top- k most relevant items of u based on the ground truth. Two issues arise from the above formulation: (i) computing user utility based on predicted relevance $\hat{r}_{u,i}$ may not reflect the true recommendation effectiveness; and (ii) relevance ties for items in R_u^k are not handled, leading to multiple possible compositions of R_u^k . Further, in [1, 7], the predicted relevance score is also taken from a latent factorisation model, which means that recommending items based on decreasing predicted relevance (as commonly done), will always result in $\phi_u^{ori}(L_u) = 1$ for all users, rendering the user utility measure pointless. We resolve the above issues

in the utility function by replacing the predicted relevance with ground truth relevance $r_{u,i}$, (e.g., ratings in the test set) as follows:

$$\phi_u(L_{u'}) = \frac{\sum_{i \in L_{u'}} r_{u,i}}{\sum_{i \in R_{u'}^k} r_{u,i}} \quad (2)$$

For binary relevance, $r_{u,i} \in \{0, 1\}$, assuming that each user has at least k relevant items, $\phi_u(L_u) = \frac{1}{k} \sum_{i \in L_u} 1_{\{i \in R_u\}}$ or simply Precision@ k (P@ k). The indicator function $1_{\{\cdot\}}$ returns 1 if the expression $\cdot$ is true, and 0 otherwise. We denote the set of user u 's relevant items as R_u . As such, the SD based on the reformulated utility function is SD-P = $SD_{u \in U} \phi_u(L_u)$, where U is the set of users.

A.2 Gini Index (Gini)

Gini is originally a measure for economic inequality [2, 5]. To measure individual fairness for users, Gini is computed as follows [4, 6]:

$$Gini = \frac{\sum_{j=1}^N (2j - N - 1)x_j}{N \sum_{j=1}^N x_j} \quad (3)$$

where $x_1, x_2, \dots, x_m$ are the recommendation EFF scores (e.g., Precision@ k or NDCG@ k) for m users.

A.3 Envy-based measures

Envy is defined as the extra utility a user u would have received if they were given the recommendation list of another user u' , $L_{u'}$ [1, 7]. Formally, the envy of user u towards u' is computed as:³

$$envy(u, u') = \max\{\phi_u(L_{u'}) - \phi_u(L_u), 0\} \quad (4)$$

There are three envy-based individual user fairness evaluation measures. Each aggregates envy differently: Mean Envy ($\downarrow$ ME) [1, 7], Mean Max Envy ($\downarrow$ MME) [3], and Proportion of ϵ -Envious User ($\downarrow$ PEU) [3]:

$$ME = \frac{2}{m(m-1)} \sum_{u \in U} \sum_{\substack{u' \in U \\ u' \neq u}} envy(u, u') \quad (5)$$

$$MME = \frac{1}{m} \sum_{u \in U} \max_{u' \in U} \{envy(u, u')\} \quad (6)$$

³ The original user utility function (Eq. (1)) is used to compute envy in [1, 7]. WLOG, user utility is computed slightly differently in [3] as the work is on online learning.

$$PEU = \frac{1}{m} \sum_{u \in U} 1_{\{\max_{u' \in U} \text{envy}(u, u') > \epsilon\}} \quad (7)$$

where $m = |U|$ is the number of users, and ϵ is the envy tolerance hyperparameter. A user u is said to be ϵ -envious of user u' if their gain in user utility (envy) exceeds ϵ .

A.4 Distance-based measures

In [8], fairness for individual users is defined as any two similar users u, u' receiving similar recommendations. Recommendation disparity is then measured with UnFairness score ($\downarrow$ UF). UF uses both the user similarity, $\text{sim}_{UF}(u, u')$, and the pairwise distance between the representations (e.g., embeddings) of items in the recommendation list of user u and of user u' , $d_L(u, u')$. User similarity is modelled by both Jaccard (sim_{Jacc}) of the user's set of past interactions H_u (Eq. (11)) and JS-div between item feature (e.g., genre) distributions in the interactions of user u and u' , $D_{JS}(u_\alpha, u'_\alpha)$, where α is the item feature. UF is computed as:

$$UF = \log \sum_{(u, u') \in U^*} \text{sim}_{UF}(u, u') \times d_L(u, u') \quad (8)$$

$$\text{sim}_{UF}(u, u') = \gamma \text{sim}_{Jacc}(u, u') + (1 - \gamma) \text{sim}_{JS}(u, u') \quad (9)$$

$$d_L(u, u') = \frac{1}{k^2} \sum_{i \in L_u} \sum_{i' \in L_{u'}} d_{cos}(\vec{i}, \vec{i'}) \quad (10)$$

$$\text{sim}_{Jacc}(u, u') = \frac{|H_u \cap H_{u'}|}{|H_u \cup H_{u'}|} \quad (11)$$

$$\text{sim}_{JS}(u, u') = 1 - D_{JS}(u_\alpha, u'_\alpha) \quad (12)$$

where U^* is the set of similar user pairs (u, u') with $\text{sim}_{UF}(u, u') \geq t$, and t is the similarity threshold. In [8], $t = \text{mean}(\text{sim}_{UF}) + SD(\text{sim}_{UF})$, where the mean and SD are computed for all user pairs.⁴ We denote as γ the weight for sim_{Jacc} and as $d_{cos}(\vec{i}, \vec{i'})$ the cosine distance between the representations of items i and i' . The log base of Eq. (8) in [8] is unclear.

B Extended Results

B.1 Varying relevance score distribution

We present the results of artificially changing the relevance score distribution for Lastfm and ML-10M in Fig. 1.

B.2 Varying user similarity distribution

We present the results of artificially changing the user similarity distribution for Lastfm and ML-10M in Fig. 2.

⁴ We also use the same threshold in our experiments.

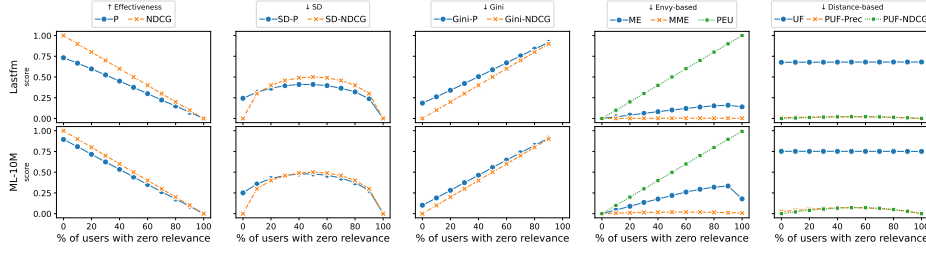


Fig. 1. Effectiveness (EFF) and fairness (FAIR) scores of Lastfm and ML-10M, when artificially varying % of users with all irrelevant items (zero relevance), and the rest of the users receiving all relevant items. All PUF variants overlap. Gini is missing points at 100% users with zero relevance as it is undefined when each user has zero EFF scores.

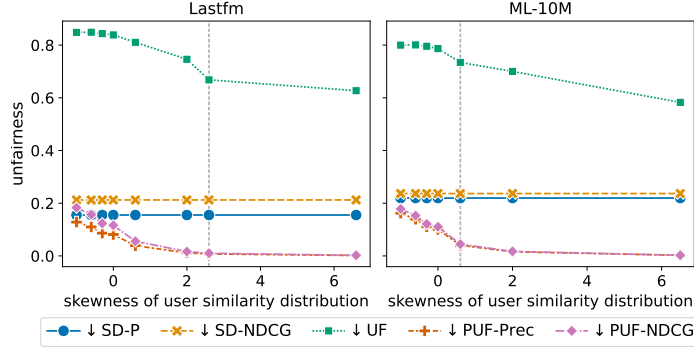


Fig. 2. Artificially varying the skewness of the user similarity distribution for Lastfm and ML-10M. Vertical grey lines denote the skewness corresponding to sim_{Jacc} observed in the dataset. The distribution skewness differs across datasets.

B.3 PUF and UF under extreme cases

We present the results of computing PUF and UF under extreme cases for Lastfm and ML-10M in Fig. 3.

B.4 Using the same similarity measure for PUF and recommendation model

We ask: if PUF and the recommendation model use the same similarity measure, would the model be the fairest according to PUF? The goal is to validate whether PUF gives an advantage to models that use the same similarity function as PUF. To answer this, we use cosine similarity (sim_{cos}) for both PUF (PUF-*-Cos) and the U-KNN model. We find that U-KNN is never the fairest according to PUF. Even though U-KNN is the second-best model based on PUF for Lastfm, it is the second worst for ML-10M and it is the unfairer for ML-20M. So, using the same similarity measure for both PUF and the recommender model does not

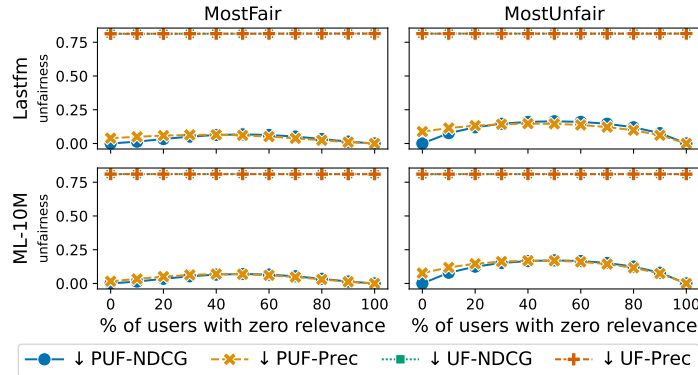


Fig. 3. Artificially varying the % of users with zero relevance for Lastfm and ML-10M. Lower EFF score difference is assigned to user pairs with higher similarity (MostFair), and to lower similarity (MostUnfair). Both UF overlap.

lead to the model being the fairest based on PUF. This may be due to the data splits: let two users have identical items in the train set, but completely different items in the test set. Even if both users get the same recommendation list, their effectiveness scores may differ and PUF cannot be the fairest. Therefore, minimising PUF to optimise for pairwise user fairness while maintaining an overall high mean effectiveness scores for all users, is not a trivial problem.

B.5 Using item features for PUF and UF

We next study what happens in a scenario where item features are available. PUF and UF are the only measures that can incorporate item features in their similarity functions. We compare PUF to UF when varying the importance of item features to them.

Procedure. We compute PUF and UF with sim_{UF} (Eq. (9)) and vary the weighted sum of its two components (Eq. (11)–(12)) with $\gamma \in \{0, 0.25, 0.5, 0.75, 1\}$. The weight $\gamma = 0$ accounts only for users’ item feature distribution, while $\gamma = 1$ accounts only for their past interactions. We evaluate the groundwork runs and the artificial runs with various relevance score distributions (§4.4) for the ML-* datasets, with movie genres as features. We cannot use the other datasets as QK-video only has two item categories and Lastfm has none.

Results. We present the results for the groundwork runs in Fig. 4 and for the artificial relevance score distributions in Fig. 5. Overall, we find that PUF scores are fairer with higher weights of the users’ past interactions. Users’ past interactions tend to be more diverse than their item category distributions, leading to more dissimilar users and less weight of user effectiveness disparity, which means

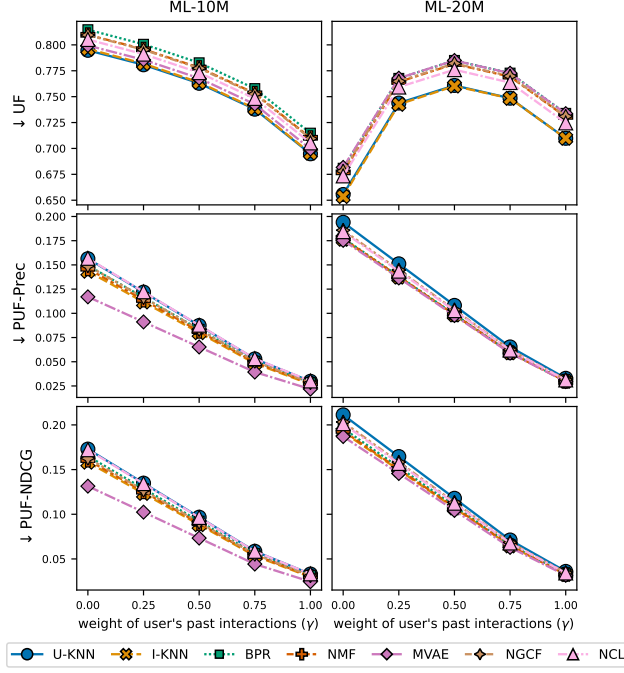


Fig. 4. UF and PUF computed with user similarity sim_{UF} (Eq. (9)), varying the weighted sum of users' past interactions (sim_{Jacc}) and their item feature distribution (sim_{JS}) with γ .

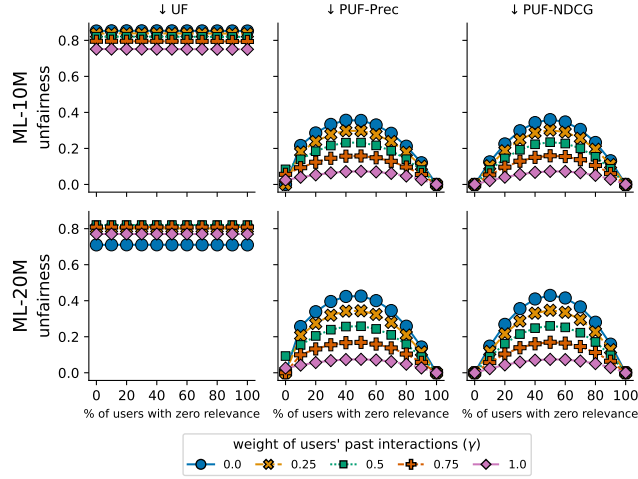


Fig. 5. Varying the weighted sum of users' past interactions (sim_{Jacc}) and their item feature distribution (sim_{JS}) with γ , while artificially varying the % of users with zero relevance (x -axis).

fairer $\downarrow$ PUF. UF follows the trend of PUF for ML-10M, but differs for ML-20M, where at first it becomes more unfair, then fairer with a higher γ . The non-linear trend of UF for ML-20M is due to its self-imposed thresholding, which results in highly different numbers of similar user pairs across various γ . While the effect of γ on UF can be inconsistent for different datasets, it is more predictable for PUF, which always exhibits near-linear relationship.

For both datasets, UF is always more unfair than PUF, which is consistent with our prior results (§4.2, §4.4–§4.6). For each measure, the ranking of models given for a weight γ is maintained for another, i.e., the same conclusion of which model is better than another can be reached despite the score differences for different weights.

To sum up, varying the weights of interaction- and feature-based user similarity impacts PUF in a more predictable way than UF.

C Extended Discussions

Having a measure that accounts for both user similarity and effectiveness disparity is not trivial. For example, dividing the user similarity by the effectiveness disparity may result in an infinitely large score if the effectiveness disparity is small, and the score will be undefined if the recommendation effectiveness is perfectly equal.

PUF captures aspects that are not accounted for in existing measures. Firstly, the non-similarity-based measures (e.g., SD) are insensitive to changes in user similarity distributions, which is a critical flaw, as they cannot give a distinct fairness score for different levels of user similarity. Secondly, the only existing measure that factors in user similarity, UF, fails to score correspondingly to the disparity in user utility and also fails to quantify contrasting fairness levels. PUF does not suffer from all these issues.

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